

src/app/codeCityLayout.ts

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Overview

`src/app/codeCityLayout.ts` is a new module that implements a layout engine for the CodeCity visualizer. It exports four public interfaces (`CodeCityFilters` , `CodeCityRenderableDistrict` , `CodeCityRenderableBuilding` , `CodeCityLayoutResult`) and a `buildCodeCityLayout` function that converts a `CodeCityModel` into renderable districts and buildings.

Key changes

- **Imports** (lines 1-8): pulls `CodeCityChangeState` , `CodeCityFileNode` , `CodeCityModel` , `CodeCityOwnershipSummary` , `CodeCityRootSide` , `CodeCitySymbolNode` from `./types` .
- **Interfaces** (lines 10-55): define filter options and the shape of the layout output.
- **Utility functions** (lines 68-106):
 - `hashColor` – deterministic hash to RGB.
 - `kindColor` – maps symbol kind to a base color.
 - `changeEmissive` – maps change state to an emissive value.
 - `buildingHeight` – computes height from line count.
 - `visibleForRoot` – determines visibility based on root side and overlay flags.
- **`buildCodeCityLayout`** (lines 129-283):
 - Returns empty arrays if `model` is `null` .
 - Builds a file map, filters symbols according to `CodeCityFilters` , groups them by district and file, and calculates 3-D coordinates for districts and buildings.
 - Sets dimensions, colors, emissive values, and ghost status based on the symbol's properties and the filter settings.

Impact

- The module is self-contained; it does not modify existing code.
- All new types and the layout function are exported, keeping the public surface explicit.
- The implementation follows a single pass over symbols and files, yielding deterministic output.

Risks & follow-ups

- **Type compatibility:** ensure `CodeCitySymbolNode` and `CodeCityFileNode` expose the properties used (`rootSide` , `changeState` , `lineCount` , etc.).
- **Rendering assumptions:** verify that the constants (`BUILDING_STEP` , `FILE_BLOCK_SIZE`) match the renderer's expectations.
- **Test coverage:** add unit tests for `buildCodeCityLayout` to guard against regressions in filtering logic.

- **Linting & build:** run `npm run lint`, `npm test`, and `npm run build` to confirm the new file passes all checks.